Exact quantum dynamics of open systems from memory kernel coupling theory: computation of higher-order moments

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Correlation Function and Memory Kernel

Background: Auto-correlation function $C_{AA}(t)$ can be described by the Generalized Quantum Master Equation (GQME):

$$\frac{d}{dt}C_{AA}(t) = \Omega C_{AA}(t) + \int_0^t K(t-\tau)C_{AA}(\tau)\,\mathrm{d}\tau\,, \tag{1}$$

- GQME's Advantage: $K(t)$ typically decays much faster than $C_{AA}(t)$.

However, $K(t) \equiv (\mathrm{i}\mathcal{L}e^{\mathcal{Q}\mathrm{i}\mathcal{L}t}\mathcal{Q}\mathrm{i}\mathcal{L}A, A)(A,A)^{-1}$ itself is challenging to evaluate due to the projected propagator $\exp(\mathcal{Q}\mathrm{i}\mathcal{L}t)$. State-of-the-art solvers include:

- Exact: HEOM, QUAPI, RT-PIMC, Tensor-Train Thermo-Field.
- Approximation: Mixed Quantum Classical Trajectories, LSC-IVR.

Memory Kernel Coupling Theory (MKCT) [1]:

- Define the higher-order moments Ω_n and memory kernels $K_n(t)$

$$\Omega_n = ((\mathrm{i}\mathcal{L})^n A, A)(A,A)^{-1}, \tag{2}$$

$$K_n(t) = ((\mathrm{i}\mathcal{L})^n e^{\mathcal{Q}\mathrm{i}\mathcal{L}t}\mathcal{Q}\mathrm{i}\mathcal{L}A, A)(A,A)^{-1}. \tag{3}$$

- Kernels $K_n(t)$ satisfy the following *initial value problem*:

$$\frac{\mathrm{d}}{\mathrm{d}t}K_n(t) = K_{n+1}(t) - \Omega_n K_1(t), \quad K_n(0) = \Omega_{n+1} - \Omega_n\Omega_1. \tag{4}$$

- Truncation of at the N-th order via the Padé approximant of $K_{N+1}(t)$ with *only* the information of $\{\Omega_n, n = 1, \dots, N+M_1+M_2+2\}$. [We define $\tilde{\Omega}_m \equiv (\mathrm{i}\mathcal{L}(\mathcal{Q}\mathrm{i}\mathcal{L})^{m-1}\mathrm{i}\mathcal{L}A, A)(A,A)^{-1}$]:

$$K_{N+1}(t) \overset{\text{Taylor}}{\approx} \sum_{n=0}^{M} \frac{K_{N+1}^{(n)}}{n!}t^n \overset{\text{Padé}}{\approx} \frac{P_{M_1}(t)}{Q_{M_2}(t)}, \tag{5}$$

$$K_n^{(m)}(0) = \Omega_{m+n+1} - \sum_{k=0}^{m}\Omega_{n+k}\tilde{\Omega}_{m-k}, \tag{6}$$

$$\tilde{\Omega}_m = \Omega_{m+1} - \sum_{k=0}^{m-1}\tilde{\Omega}_k\Omega_{m-k}. \tag{7}$$

- Computational Efficiency: Only need to evaluate a *finite* number of moments $\{\Omega_n\}$ [, with, e.g., HEOM (PRL), Exact Symbolic (This work)].

Exact Symbolically Evaluation of Ω_n

$$H = H_{\text{sys}} + H_{\text{bath}} + \sum_{n=1}^{N} F_n Q_n \tag{8}$$

Bosonic reservoir

Condensed phase vibration (solvents or lattices)

$$F_n = \sum_j c_{nj}(b_{nj}^\dagger + b_{nj}) = \sum_j c_{nj}x_{nj}$$

Fermionic reservoir

A continuum of electronic states (electrodes)

$$F_n = \sum_j V_{nj}c_{nj}, \quad F_n^\dagger = \sum_j V_{nj}c_{nj}^\dagger$$

The exact commutator hierarchies

$$(\mathrm{i}\mathcal{L})^n A = \sum_t^{\text{all terms}} O_t \prod_{n=1}^{N}\left[\prod_{\alpha_t} X_{n\alpha_t}\prod_{\beta_t} Y_{n\beta_t}\right], \quad (X,Y) = \begin{cases}(Q,P) & \text{Boson}\\ (C^\dagger, C) & \text{Fermion}\end{cases}.$$

Gen. coords.: $Q_{n\alpha} = \sum_j c_{nj}\omega_{nj}^\alpha x_{nj}$,

Gen. mom.: $P_{n\beta} = \sum_j c_{nj}\omega_{nj}^\beta p_{nj}$.

Gen. cre. $C_{n\alpha} \equiv \sum_j \epsilon_{nj}^\alpha V_{nj}c_{nj}$,

Gen. ann. $C_{n\alpha}^\dagger \equiv \sum_j \epsilon_{nj}^\alpha V_{nj}c_{nj}^\dagger$

*For equations of $\mathrm{i}\mathcal{L}_{\text{sys}}$, $\mathrm{i}\mathcal{L}_{\text{bath}}$, and $\mathrm{i}\mathcal{L}_{\text{sys-bath}}$, see our notes (fermionic case) and the published paper (bosonic case).

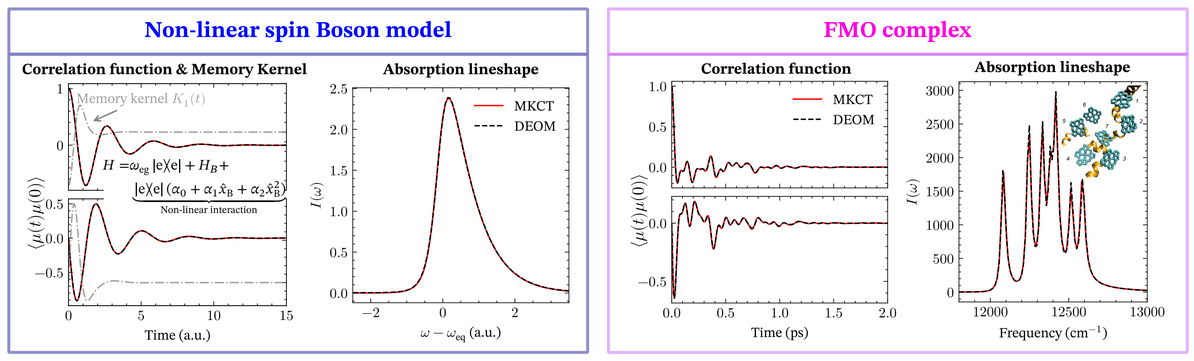
Evaluation of Ω_n cont.

Moments from commutation expressions:

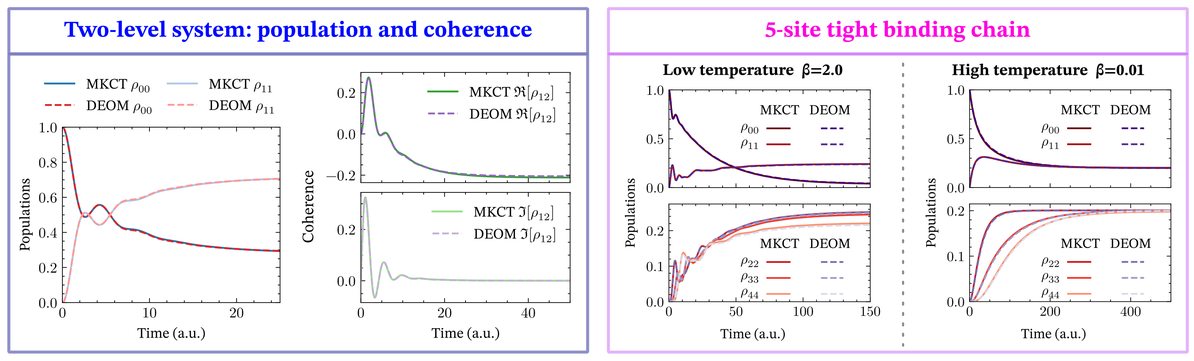
$$\Omega_n \equiv \frac{((\mathrm{i}\mathcal{L})^n A, A)}{(A,A)}, \quad \rho(0) = \sigma(0)\otimes\rho_{\text{eq}}^{\text{B}} \quad \Longleftarrow \quad \frac{\mathrm{Tr}[O_t A\sigma(0)]}{\mathrm{Tr}[AA\sigma(0)]}\prod_n^N\left\{\mathrm{Tr}\left[\prod_{\alpha_t}X_{n\alpha_t}\prod_{\beta_t}Y_{n\beta_t}\rho_{\text{eq}}^{\text{B}}\right]\right\}$$

Results and Discussion

Linear Spectroscopy:



Reduced density matrix Dynamics:



Summary and Future Directions

We present a formally exact symbolic framework for computing Ω_n in open systems, unifying bosonic and fermionic dynamics. Combined with MKCT, this enables efficient calculations of

- Dipole autocorrelation for linear absorption (Bosonic).
- Population and coherence of reduced system (Bosonic).

Besides these, we have ongoing works on:

- Anderson impurity model at finite U (Fermionic).
- Dissipative Holstein model.

Paper and Code

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References

[1] W. Liu, Y. Su, Y. Wang, and W. Dou, "Memory kernel coupling theory: obtaining time correlation function from higher-order moments", Phys. Rev. Lett. (accepted), 10.1103/qvd5-5z6m (2025).